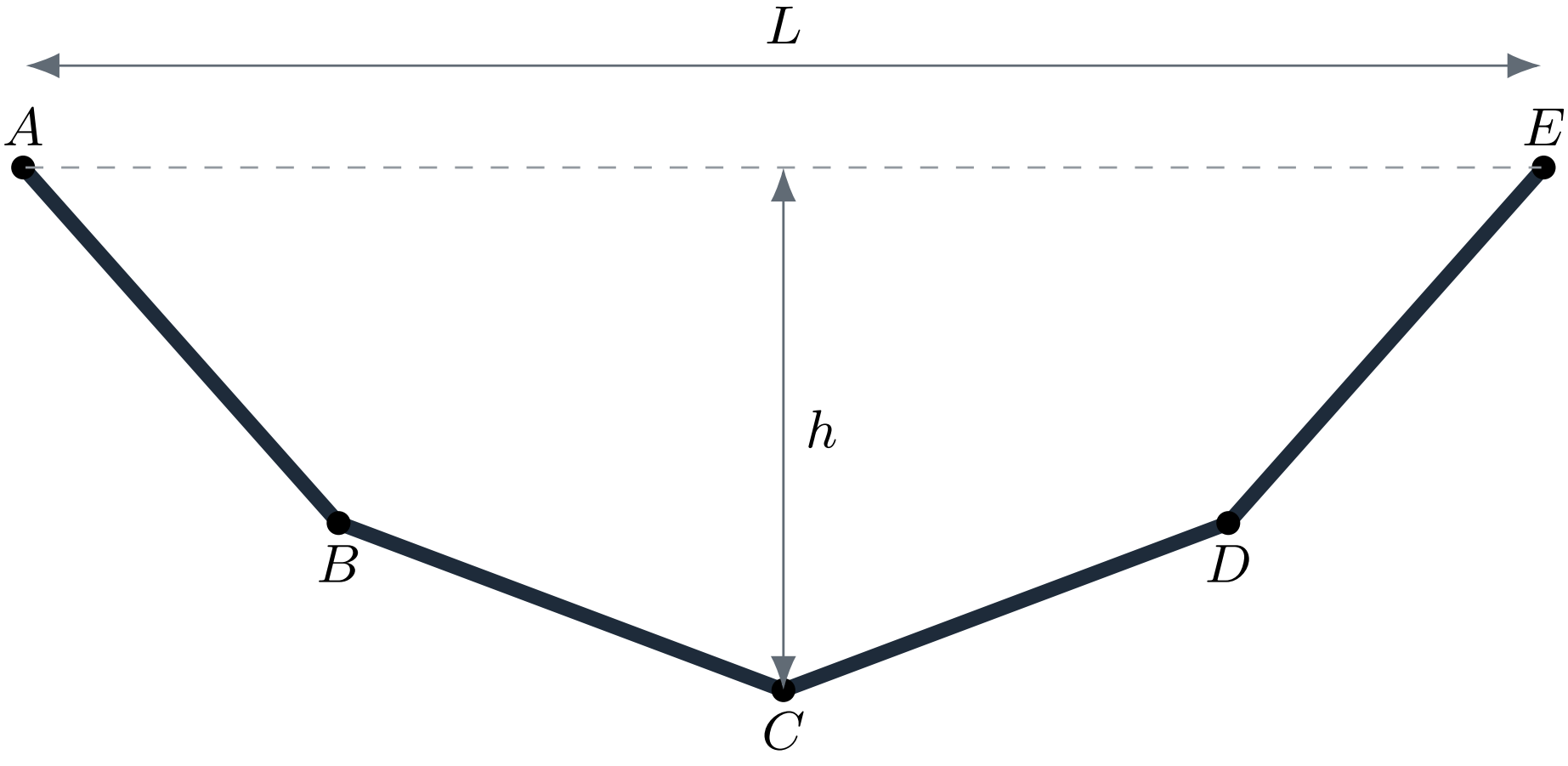
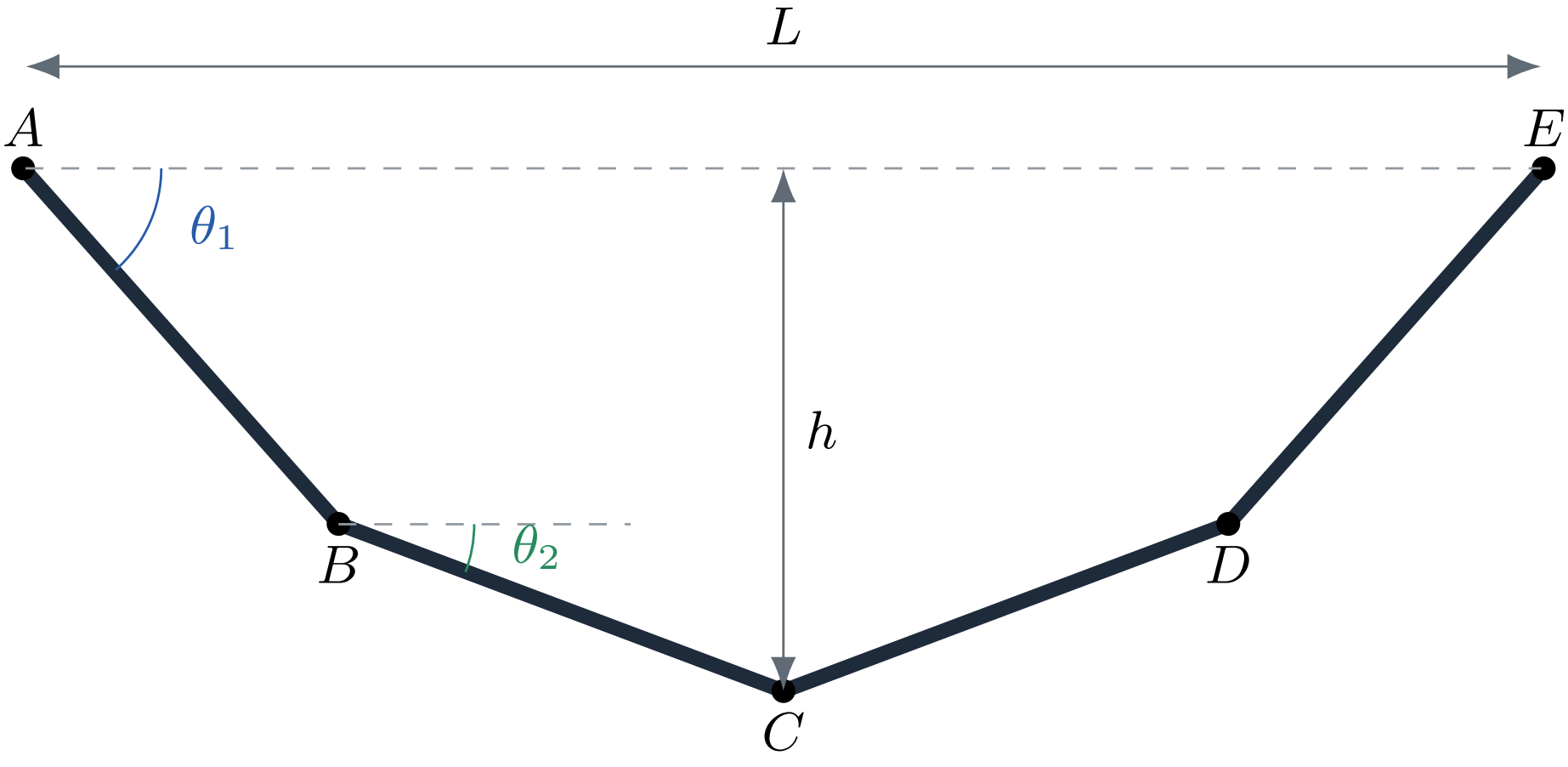
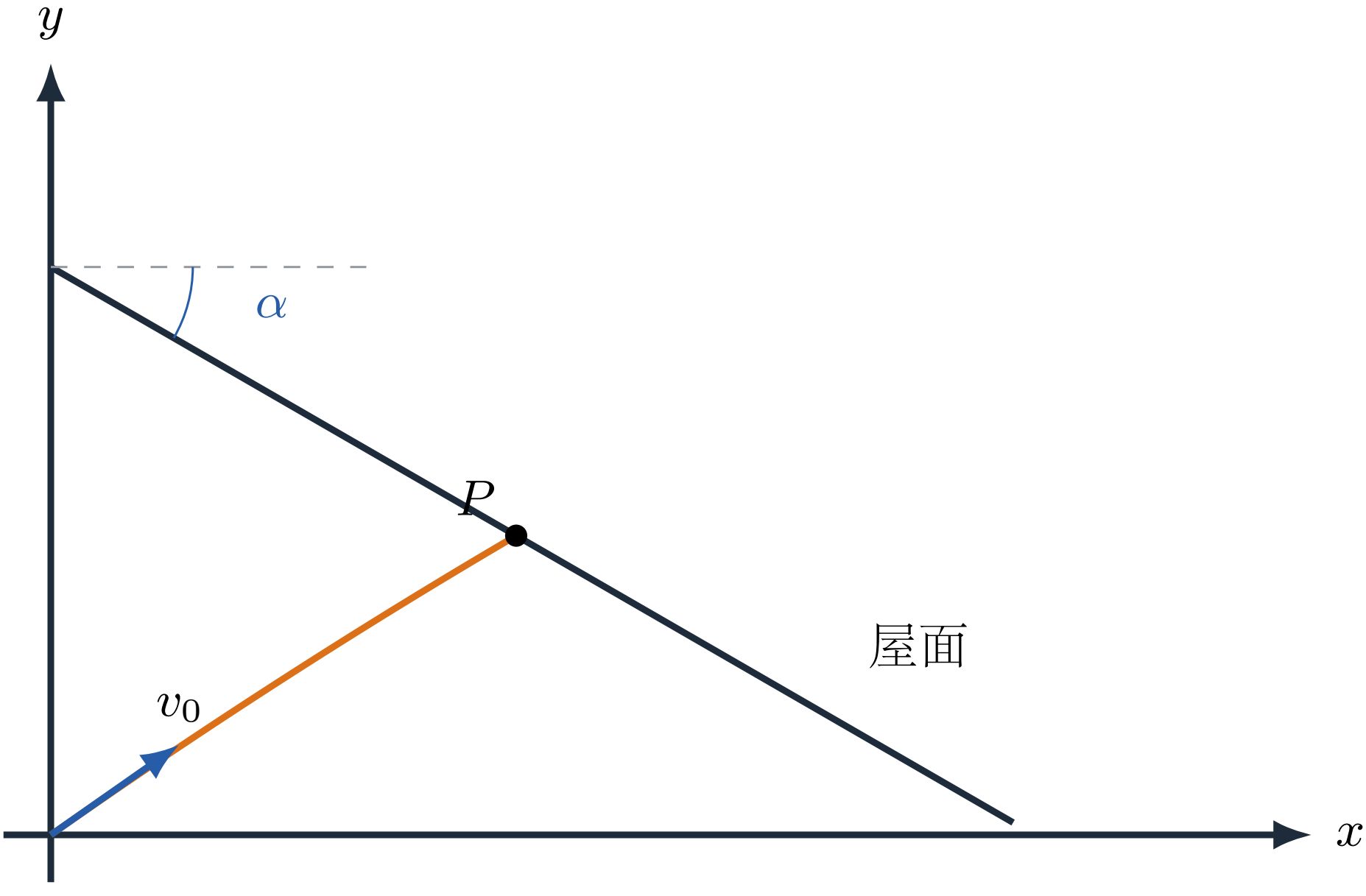
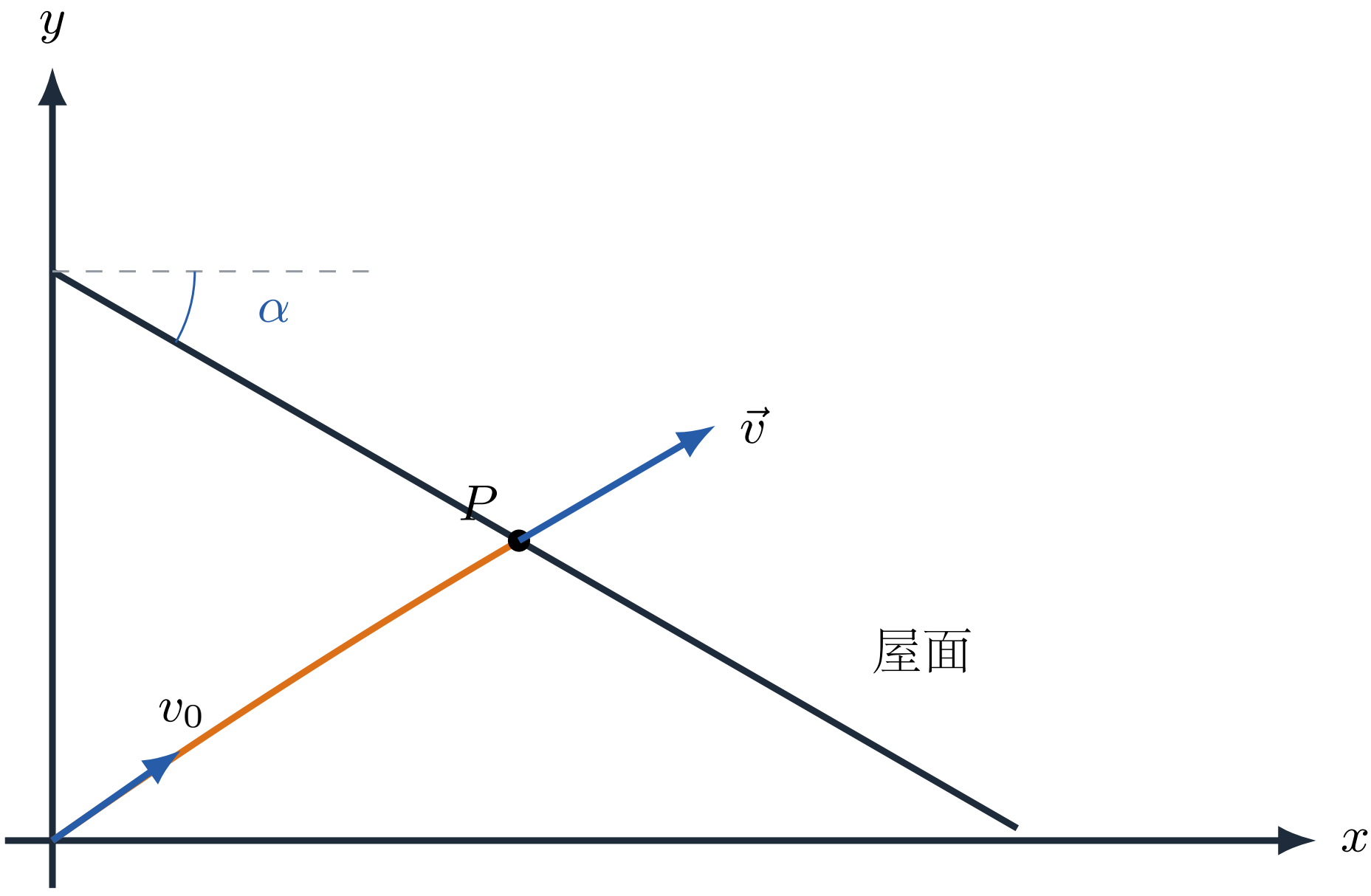

L
A
E
h
B
D
C



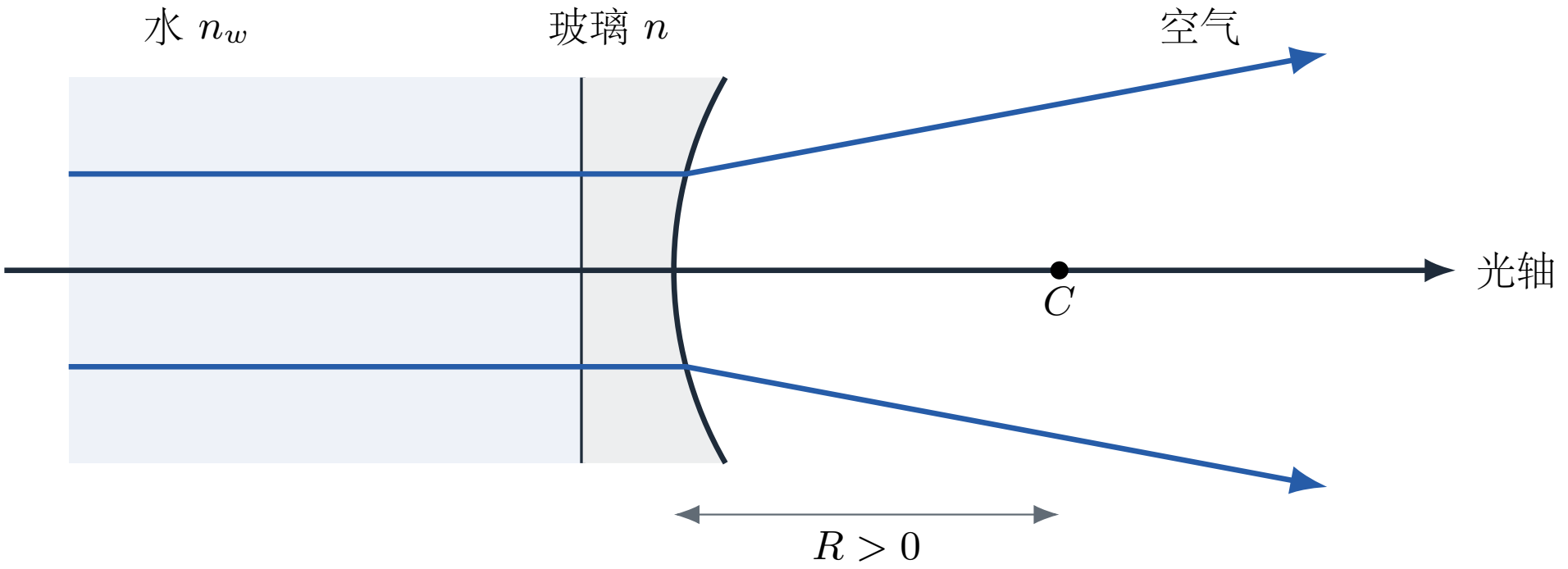
$$\tan\theta_1 = 3\tan\theta_2$$

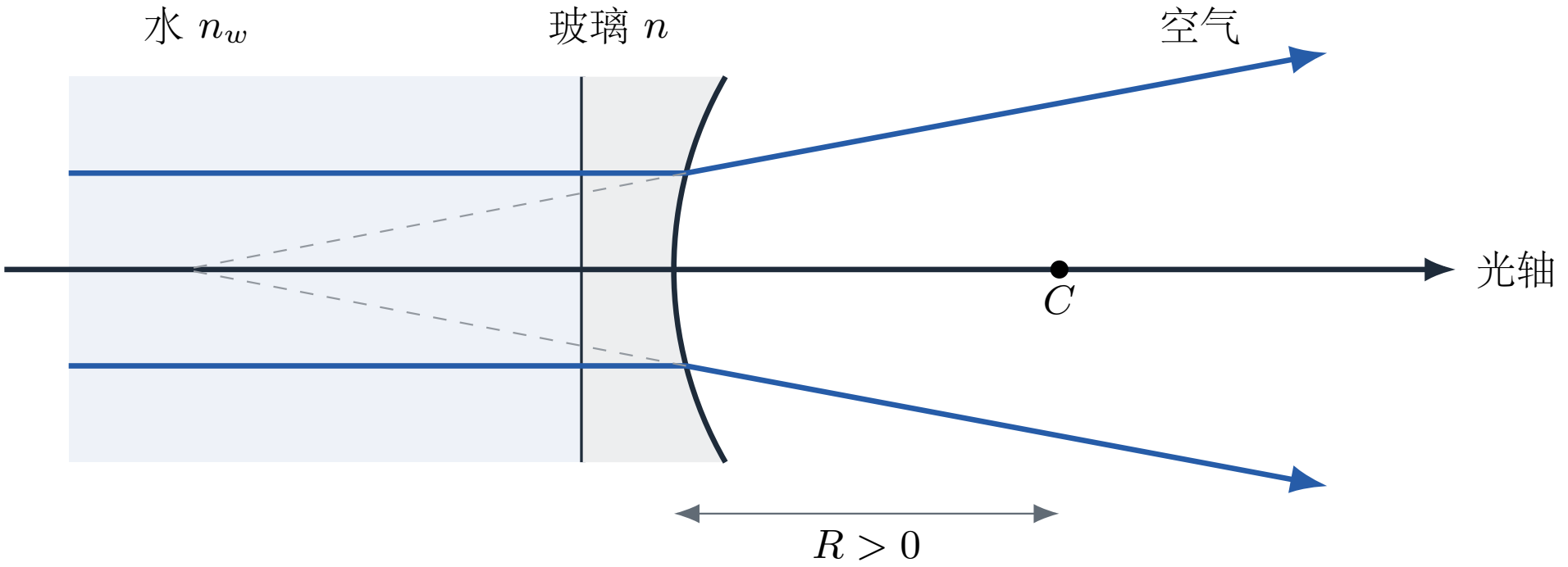

y
α
P
屋面
v_0
x



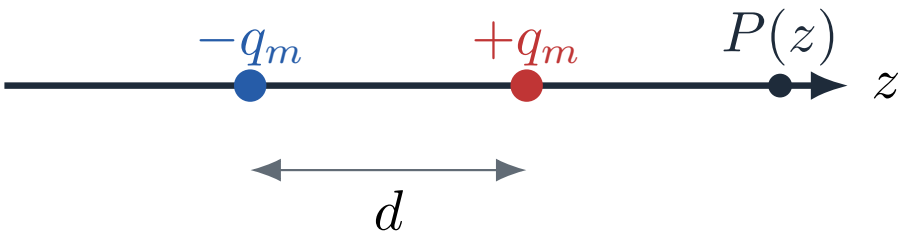
$$\theta_{\text{opt}} = \pi/2 - \alpha$$

$$y = h - x\tan\alpha$$

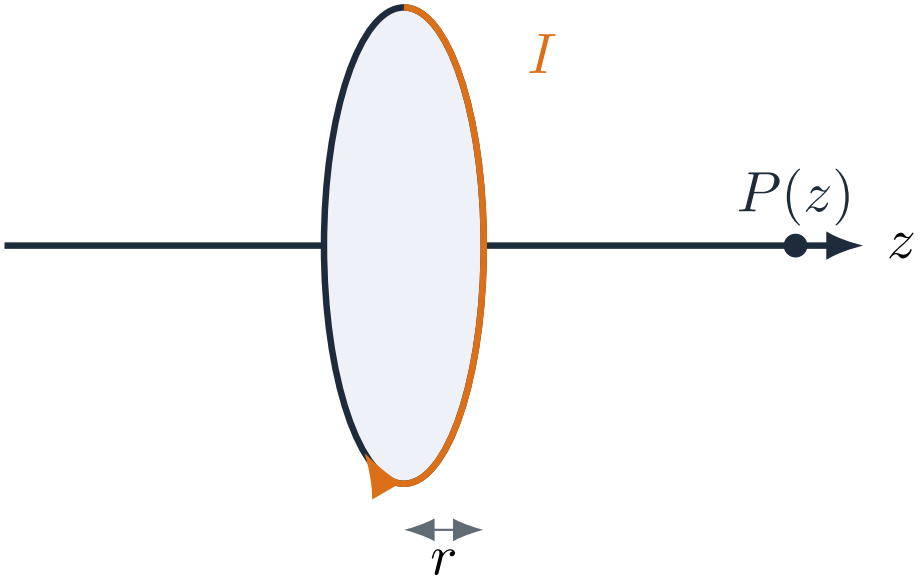

水 n_w
玻璃 n
空气
光轴
C
$R > 0$



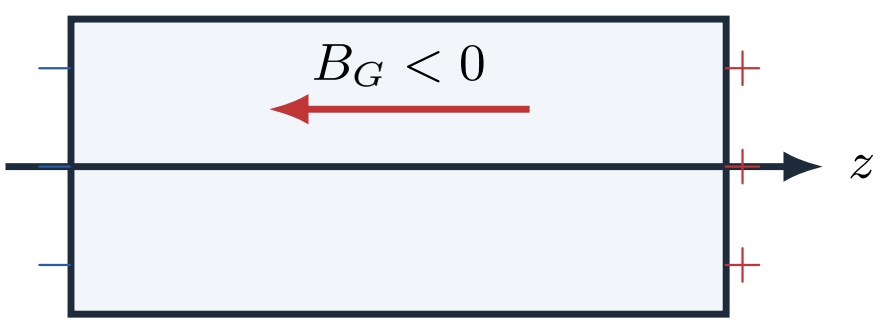
$\Phi = (1-n)/R < 0$，反向延长线给虚焦点

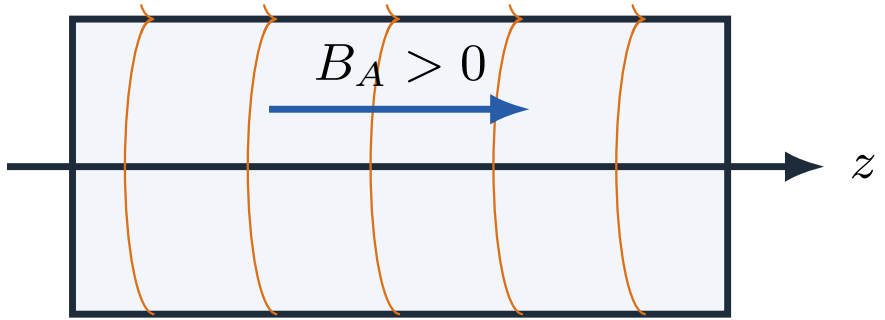


Gilbert 模型



Ampere 模型





$$B_G = -\mu_0 M(1 - a/\sqrt{a^2 + R^2})$$

$$L = 2a$$

$$B_A = \mu_0 M a/\sqrt{a^2 + R^2}$$

$$L = 2a$$

$B_A - B_G = \mu_0 M$；辅助磁荷算的是 H

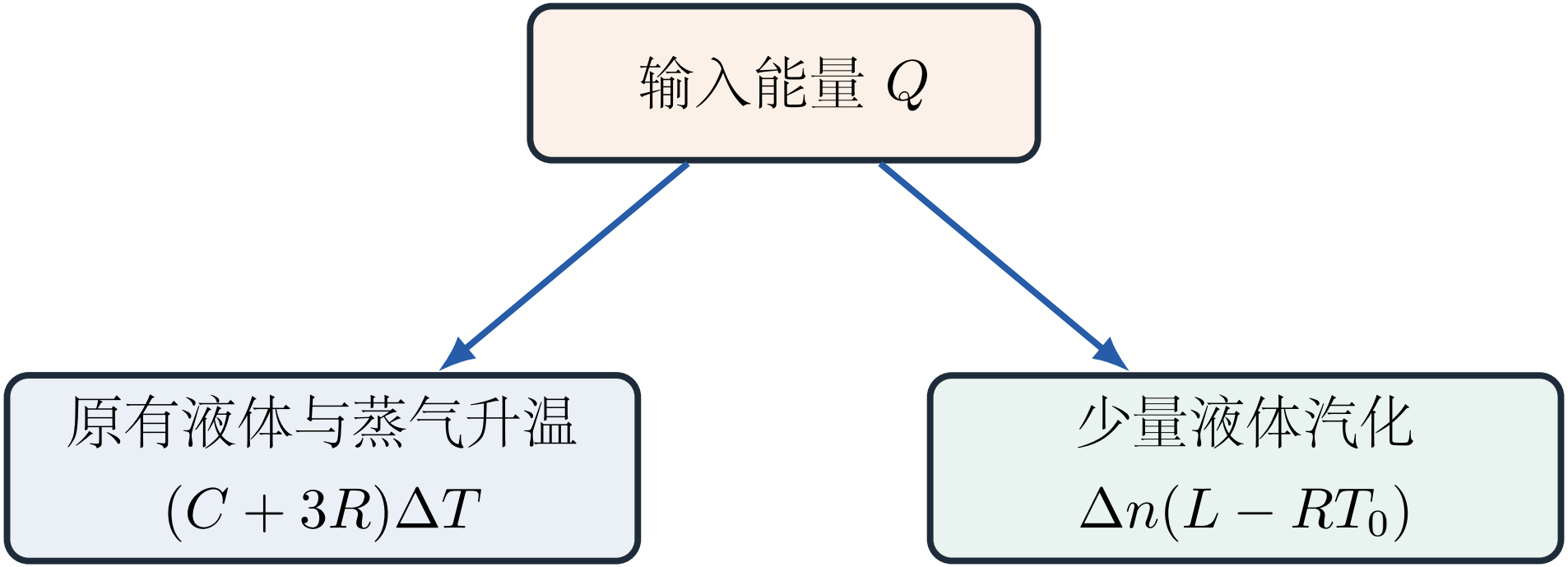
器壁绝热，V = 常量

饱和水蒸气 1 mol



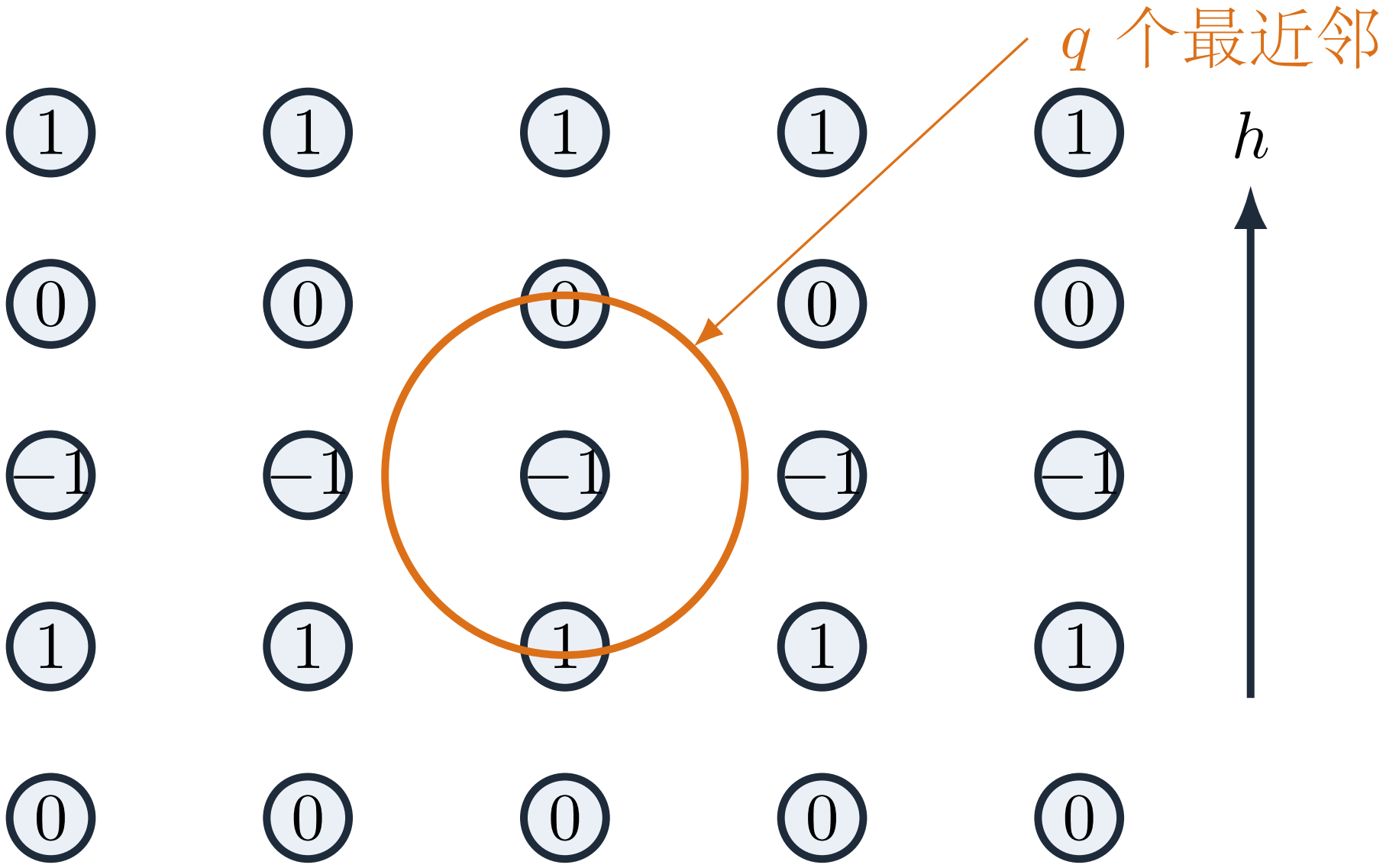
电热丝

液态水 1 mol

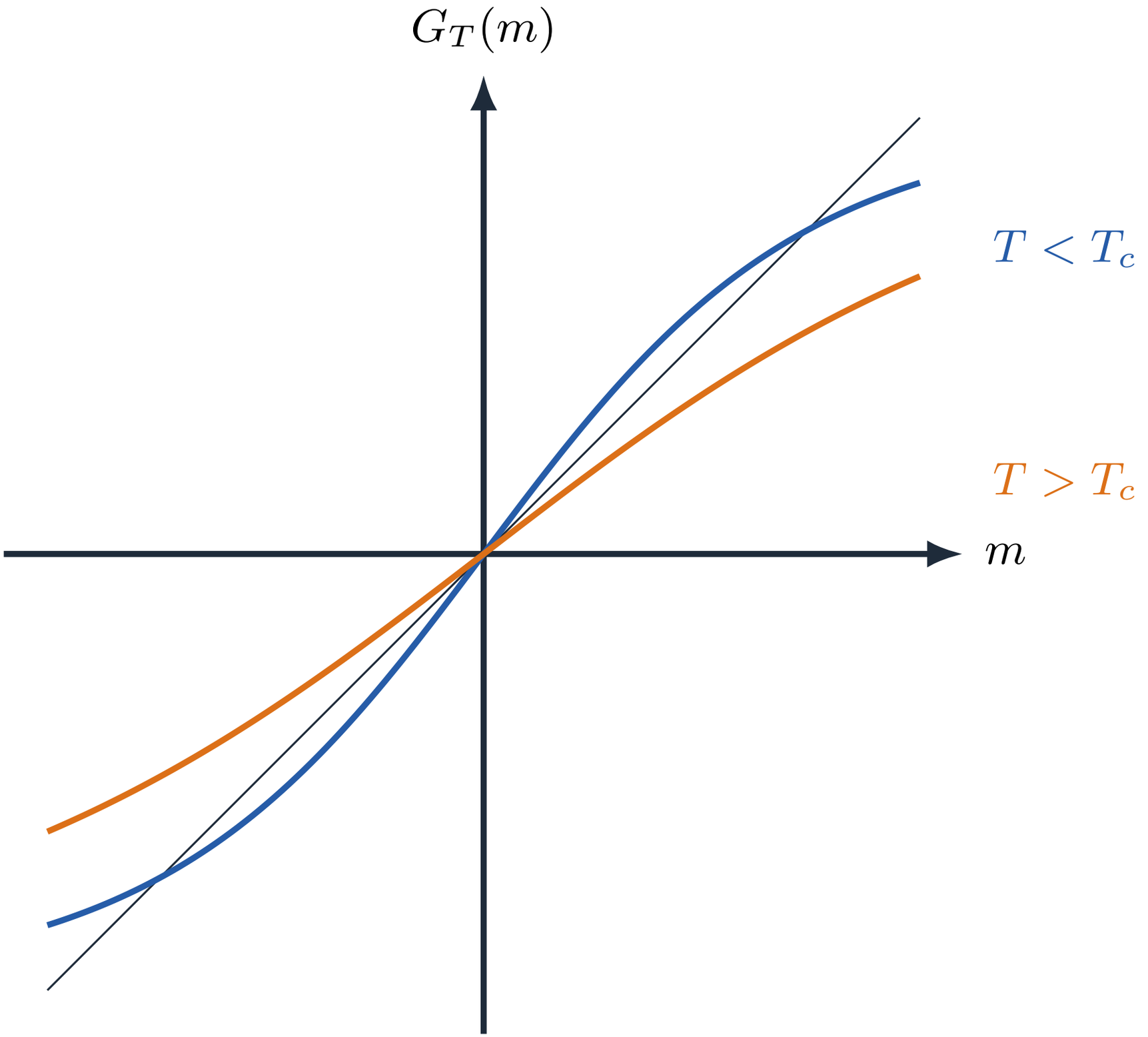


$$\Delta n = \frac{n_0 \Delta T}{T_0}\left(\frac{L}{RT_0} - 1\right)$$

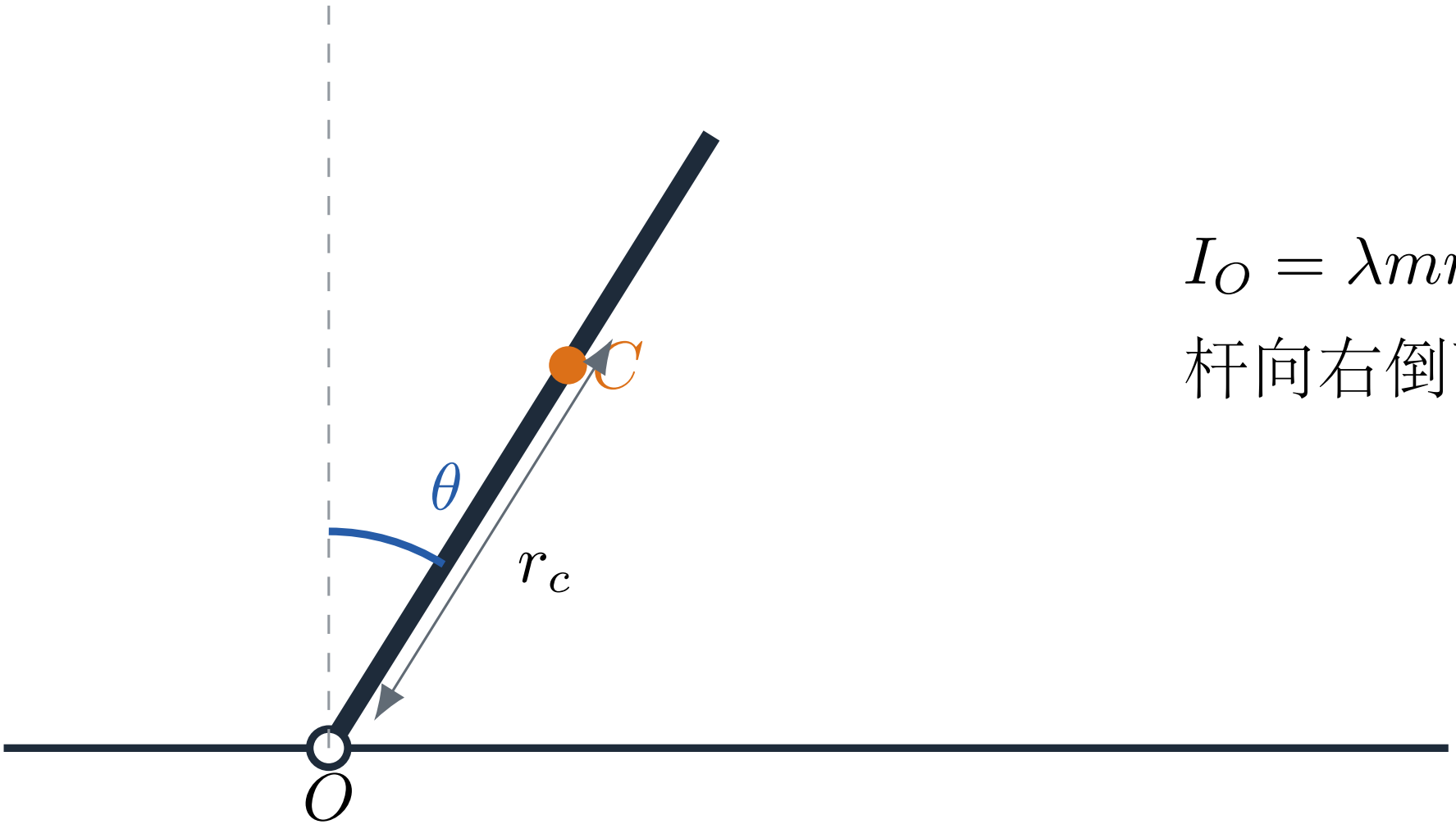
$$Q = (C+3R)\Delta T + \Delta n(L - RT_0)$$



$$S_z \in \{-1, 0, +1\}$$

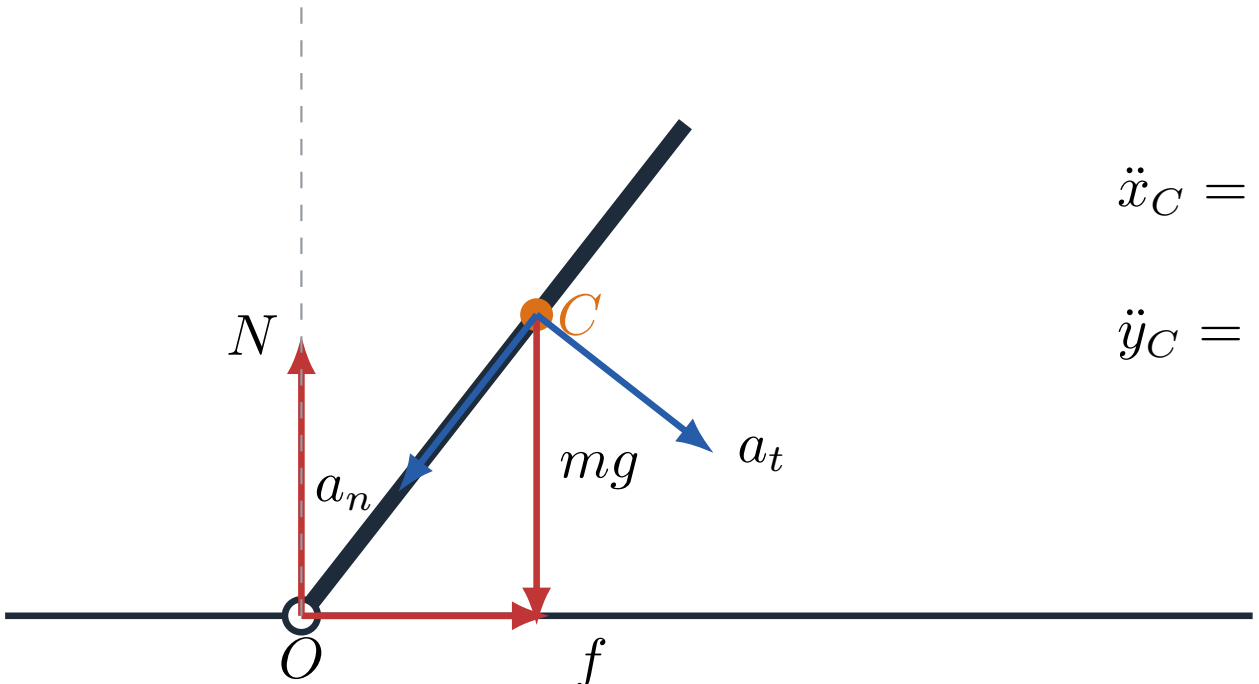


$$G_T(m) = 2\sinh y/(1+2\cosh y), \quad y = qJm/(kT)$$



$$I_O = \lambda m r_c^2$$

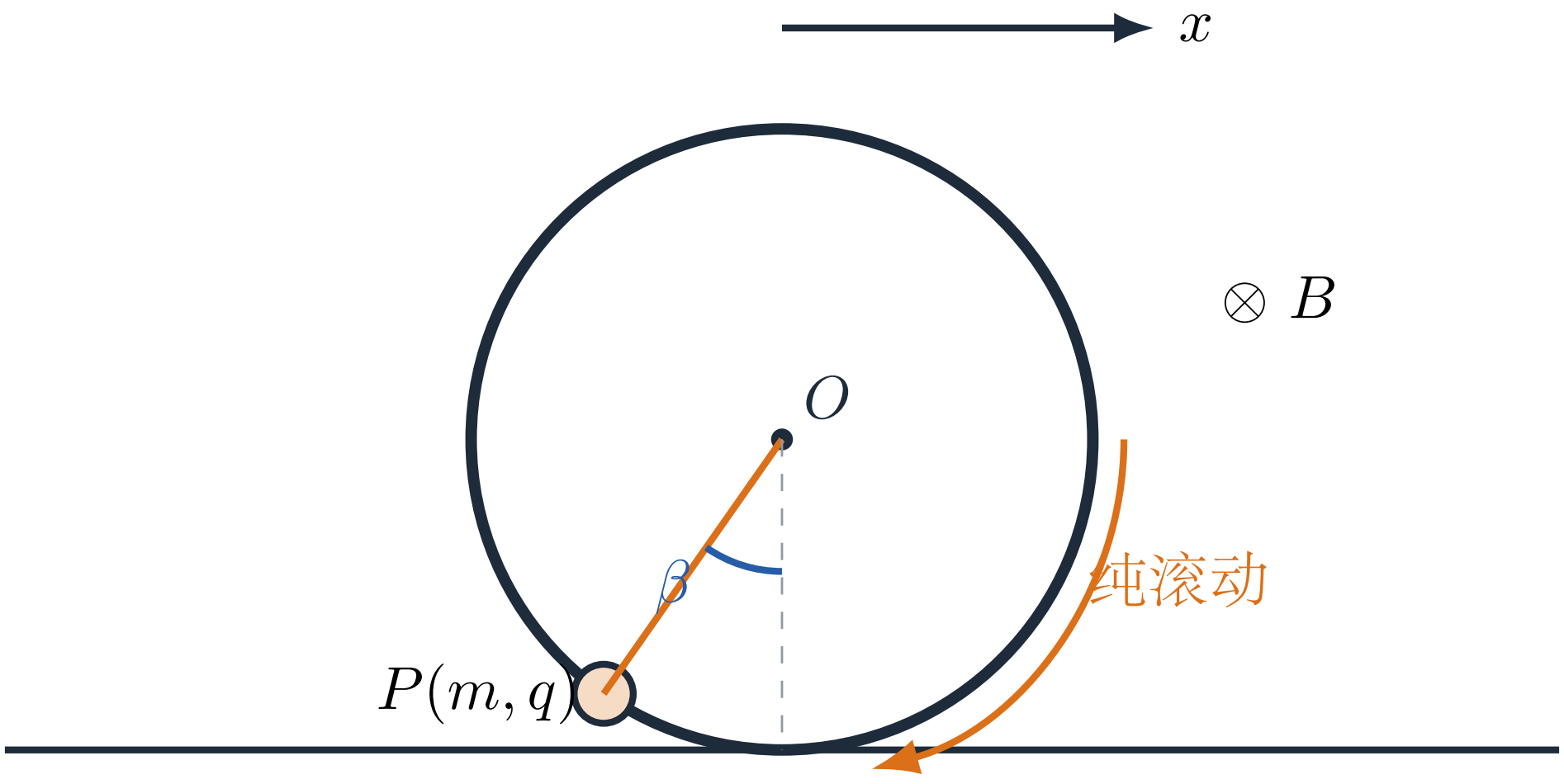
杆向右倒下



$$\ddot{x}_C = \frac{g}{\lambda}\sin\theta(3\cos\theta - 2)$$

$$\ddot{y}_C = -\frac{g}{\lambda}(1 + 2\cos\theta - 3\cos^2\theta)$$

$$a_t = r_c\ddot{\theta}, \quad a_n = r_c\dot{\theta}^2$$

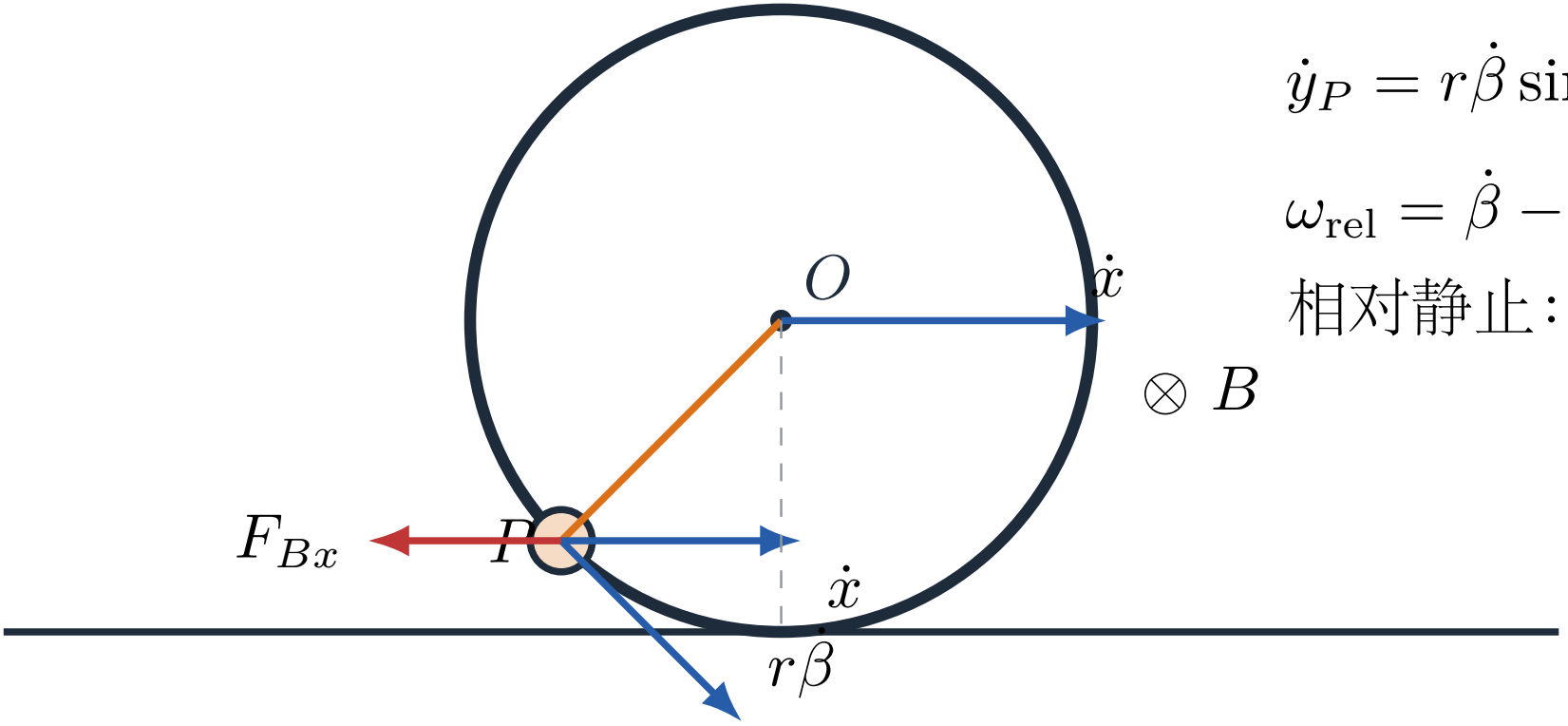

x
B
O
β
纯滚动
P(m, q)

$$\dot{x}_P = \dot{x} - r\dot{\beta}\cos\beta$$

$$\dot{y}_P = r\dot{\beta}\sin\beta$$

$$\omega_{\text{rel}} = \dot{\beta} - \dot{x}/r$$

相对静止：$r\dot{\beta} = \dot{x}$



$$3\dot{x} - r\dot{\beta}\cos\beta = (qBr/m)\cos\beta$$